

Karthik Reddy Jakka

AI / ML Engineer | Generative AI & Data Science

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SUMMARY

Final-year IT student building applied AI and data-analytics systems across generative AI, machine learning, and computer vision, and authoring agentic coding benchmarks for frontier-model evaluation as a freelance AI trainer at Handshake AI. Hands-on across Python, LLM-based conversational analysis, predictive modelling, and end-to-end research systems — including a knee-MRI diagnostic AI platform and a full-stack agricultural intelligence platform combining ML, RAG, and LLM-driven advisory. Oracle-certified in Generative AI and Data Science. Seeking AI/ML or data engineering roles building production-grade intelligent systems.

WORK EXPERIENCE

Freelance AI Trainer — Agentic Benchmark Authoring

2026 – Present

Handshake AI

Remote

- Developed terminal-based benchmark tasks to evaluate frontier AI coding agents across systems infrastructure, model-training/ML pipelines, data processing, databases, and interactive simulation domains.
- Deliver each task as a reproducible package — natural-language specification, Dockerised environment, automated test-based verifier, and a working reference solution — ensuring every component meets automated QA standards and passes human review.

Prompt Engineering for AI — Virtual Internship

Apr 2026 – Jun 2026

EduSkills Academy / AICTE-EduSkills

Remote

- Completed a 10-week virtual internship covering core prompt engineering techniques, OpenAI API prompting, Hugging Face Transformers, and prompt design for text, code, chatbots, image generation, and data analysis.
- Applied advanced prompting strategies and domain-specific techniques in a capstone AI-powered content generation and analysis platform.

AI-ML Virtual Internship

Nov 2025 – Jan 2026

Google

Remote

- Designed and implemented machine learning models using Python to perform predictive data analysis.
- Applied data preprocessing and model evaluation techniques to improve forecasting performance.

Data Science Intern

Oct 2025 – Nov 2025

Oasis Infobyte

Remote

- Conducted exploratory data analysis and built data visualizations using Python libraries to uncover key insights.
- Developed machine learning models for trend and pattern analysis, delivering measurable improvements in data insights and supporting informed decision-making.

LLM Data Analysis Intern

Sep 2025 – Oct 2025

Vodafone Idea Foundation

Remote

- Implemented LLM-based conversational data analysis workflows, improving data processing efficiency.
- Applied prompt engineering and NLP techniques to extract insights from datasets.

EDUCATION

Matrusri Engineering College — B.Tech, Information Technology

2023 – 2027

Sri Chaitanya Junior College — Intermediate, MPC

2021 – 2023

SKILLS

Hard Skills: Artificial Intelligence, Generative AI, Large Language Models (LLMs), Prompt Engineering, AI Image Generation, AI Agents, Generative AI Tools, Data Science, Data Analysis, Data Cleaning, Data Visualization, Statistical Data Analysis, Business Insights, Technical Data Analysis, Python for Data Analysis, Python, Java, C, C++, SQL, Google Analytics, Git & GitHub, VS Code, Microsoft 365

Soft Skills: Problem Solving, Technical Communication

PROJECTS

KneeVision-X — Research-Grade Knee MRI AI | *Python, PyTorch, FastAPI, React, DICOM, Multimodal AI* 2026 – Present

- Built a multi-plane, multi-pathology knee MRI AI system with Grad-CAM explainability, uncertainty estimation, calibration, and automated clinical-style reporting.
- Designed Pathology-Aware Report Contrastive Learning (PARCL) for MRI–radiology-report alignment and label-efficient multimodal representation learning.
- Integrated MRNet, SKM-TEA, fastMRI, RSNA 2026, and independent OA datasets with leakage checks, robustness testing, and reproducible evaluation.
- Achieved **0.832 macro AUROC** on knee OA severity grading and developed a full-stack FastAPI + React research dashboard.

AgriIntel AI — Intelligent Crop Yield Prediction & Smart Farming Platform | *Python, FastAPI, PyTorch, Scikit-learn, PostgreSQL/PostGIS, Redis, Docker, LLM, RAG* 2026 – Present

- Developed an AI-powered agricultural decision-support platform integrating crop recommendation, yield prediction, plant disease detection, soil/weather intelligence, fertilizer guidance, market intelligence, and conversational AI.
- Built a HistGradientBoosting yield prediction model achieving **R² 0.846**, MAE 331.2 kg/ha, and RMSE 655.7 kg/ha on an unseen 2010–2017 test set.
- Developed a ResNet-18 plant disease classification model achieving **99.55% own-validation accuracy**.
- Implemented FastAPI REST APIs, PostgreSQL/PostGIS, Redis, Docker Compose, authentication, authorization, and object-level access control for secure deployment.
- Integrated RAG, a Knowledge Graph, and LLM-based advisory generation with validation mechanisms to improve context-aware farmer recommendations.
- Performed production validation with **464 tests passed, 0 failures**, live backup/restore testing, and multi-service Docker deployment.

Intelligent Lecture Companion | *Generative AI, LLMs* 2025 – Present

- Developed an AI-powered lecture assistant leveraging LLMs to summarize course material and answer student questions, enhancing classroom engagement and accessibility.

Airbnb Booking Analysis | *Python, Machine Learning* 2025 – Present

- Analyzed booking data and built ML models for pricing trends.

Netflix Content Analysis | *Python, Data Analytics* 2025 – Present

- Analyzed **7,789 Netflix titles** to identify genre and global trends.

CERTIFICATIONS

- Oracle Cloud Infrastructure Certified Generative AI Professional — Oracle
- Oracle Cloud Infrastructure Certified Data Science Professional — Oracle
- Oracle Fusion AI Agent Studio Certified Foundations Associate — Oracle
- Prompt Engineering for AI — 10-week Virtual Internship, EduSkills Academy (04/2026 – 06/2026)
- Data Analysis Using Python — IBM
- Career Essentials in Data Analysis — Microsoft & LinkedIn
- Introduction to Generative AI — Google
- Python Programming — IIT Bombay
- GenAI Powered Data Analytics Job Simulation — Tata Group via Forage
- Data Analytics Job Simulation — Deloitte Australia via Forage
- Generative AI Mastermind — Outskill